

# Transformer Encoder for Text Classification

Idant Srivastava

April, 2026

## Contents

|          |                                            |          |
|----------|--------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                        | <b>2</b> |
| <b>2</b> | <b>Architecture</b>                        | <b>2</b> |
| 2.1      | Encoder vs. Plain MLP . . . . .            | 2        |
| 2.2      | Encoder vs. Autoencoder . . . . .          | 2        |
| <b>3</b> | <b>Training</b>                            | <b>3</b> |
| <b>4</b> | <b>Ablation Study</b>                      | <b>3</b> |
| 4.1      | Effect of Scaling . . . . .                | 4        |
| 4.2      | Effect of No Positional Encoding . . . . . | 4        |
| 4.3      | Effect of Head Count . . . . .             | 4        |
| <b>5</b> | <b>Attention Analysis</b>                  | <b>5</b> |
| <b>6</b> | <b>Failure Analysis</b>                    | <b>5</b> |

# 1 Introduction

This report documents the implementation and analysis of a Transformer Encoder applied to the AG News text classification task. The study covers architecture design, training behaviour, systematic ablation experiments, and attention visualisation. The goal is to understand the contribution of each architectural component and identify potential failure modes in the current implementation.

## 2 Architecture

The implemented model follows the standard Transformer Encoder architecture introduced by Vaswani et al. (2017). The key components are:

- **Embedding Layer.** Token IDs are projected into a 64-dimensional continuous space ( $d_{\text{model}} = 64$ ).
- **Positional Encoding.** Sinusoidal signals are added to the embeddings to inject word-order information, since self-attention is inherently permutation-invariant.
- **Multi-Head Attention (MHA).** Four parallel attention heads allow the model to attend to different representation subspaces simultaneously. Each head operates on a sub-dimension of  $d_k = 16$ .
- **Scaled Dot-Product.** Attention logits are scaled by  $1/\sqrt{d_k}$  before the softmax to prevent the dot products from growing too large and saturating the gradient.
- **Feed-Forward Network (FFN).** A two-layer MLP with ReLU activation and hidden dimension 256 is applied position-wise after MHA.
- **Global Average Pooling.** Mean pooling across the sequence dimension produces a fixed-size representation for the final linear classification head.

### 2.1 Encoder vs. Plain MLP

A plain MLP treats input features as independent or connects them through fixed global weights, making it structurally unable to model contextual relationships between words without an enormous parameter count. The Transformer Encoder addresses this with self-attention, which dynamically reweights each token’s representation based on all other tokens in the sequence. This makes it far more effective at capturing linguistic structure such as subject–verb agreement, coreference, and phrase-level semantics.

### 2.2 Encoder vs. Autoencoder

The encoder used here is designed to extract high-level features for a classification task. An autoencoder, by contrast, is an architecture that compresses input into a bottleneck and then attempts to reconstruct the original input. Having a vector per token allows the model to perform token-level tasks or apply sophisticated pooling that preserves more semantics than a fixed-size global embedding. Essentially, the Transformer provides a much richer representation because it preserves the spatial and semantic relationships of every individual word.

### 3 Training

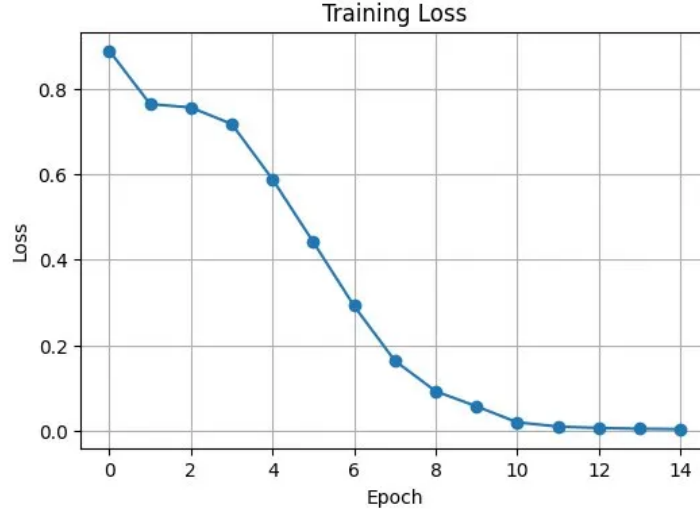


Figure 1: Baseline training loss over 15 epochs.

### 4 Ablation Study

Three ablation configurations were tested against the baseline (4 heads, with positional encoding and  $1/\sqrt{d_k}$  scaling):

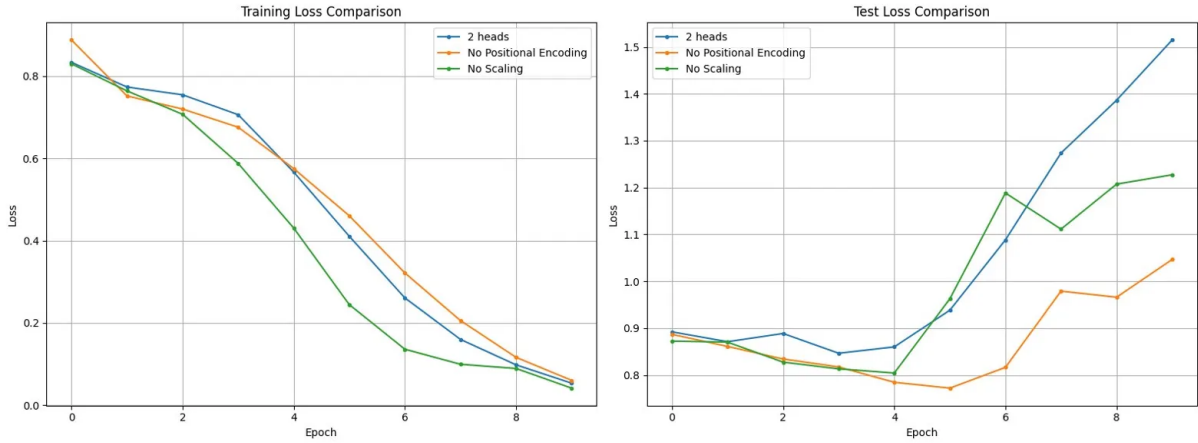


Figure 2: Training loss (left) and test loss (right) for the three ablation variants over 10 epochs.

1. **2 Heads** — reduced attention capacity.
2. **No Positional Encoding (No PE)** — model has no access to token order.
3. **No Scaling** — attention logits are not divided by  $\sqrt{d_k}$ .

## 4.1 Effect of Scaling

Without the scaling factor  $1/\sqrt{d_k}$ , the attention logits of the dot-product grow proportionally to  $d_k$ . This pushes the softmax into saturation regions where gradients are extremely small, resulting in slower and more erratic convergence. In extreme cases this can cause the model to stall entirely.

## 4.2 Effect of No Positional Encoding

Removing positional encoding converts the model into a sophisticated bag-of-words classifier. The training loss remains low because the AG News dataset contains strong class-specific keywords whose mere presence, regardless of position, is often sufficient for correct classification. Interestingly, the No-PE variant shows a lower test loss in the mid-training range compared to the 2-head baseline, suggesting that for this particular dataset, word-order sensitivity can introduce unnecessary variance.

## 4.3 Effect of Head Count

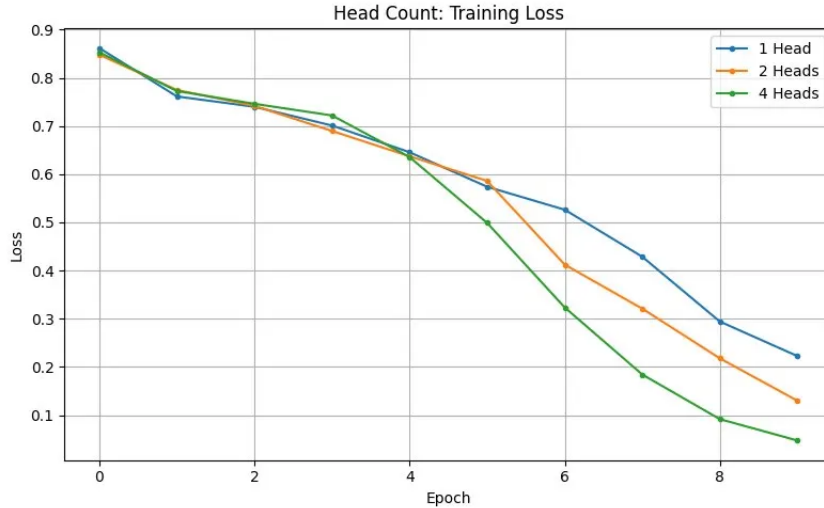


Figure 3: Training loss for 1, 2, and 4 attention heads over 10 epochs.

The graph shows that more attention heads consistently produce faster and lower training loss. This validates the benefit of parallel attention subspaces: each head can specialise in a distinct linguistic feature and their combined output produces a richer contextual representation.

## 5 Attention Analysis

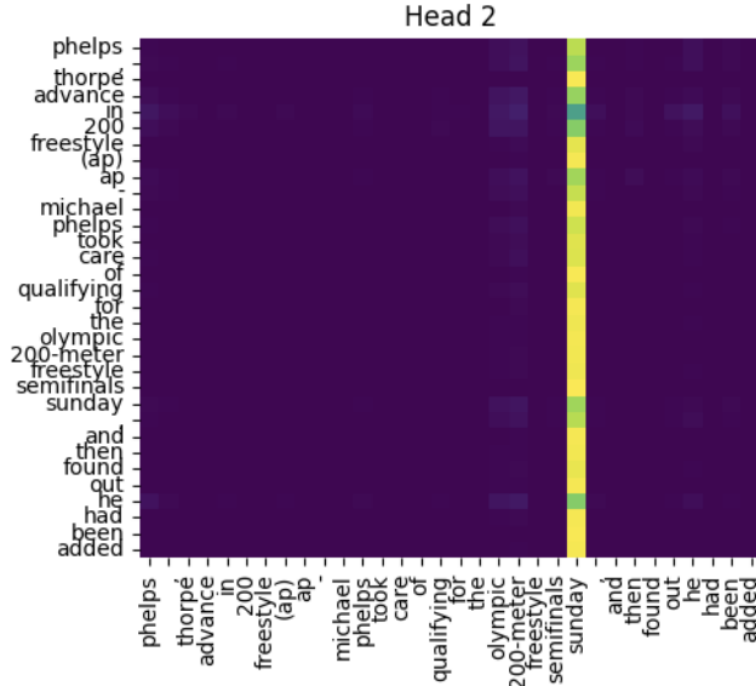


Figure 4: Attention heatmap for Head 1 on a sample sentence

This visualization shows a specific attention head where the y-axis represents the query tokens and the x-axis represents the key tokens. The vertical yellow column indicates that nearly every token is attending heavily to the word "sunday", regardless of its own position or semantic meaning. This is a highly meaningful, non-random pattern often referred to as "delimiter" attention pattern.

In many trained models, certain heads specialize in attending to specific frequent tokens, punctuation, when the head doesn't find a more relevant linguistic relationship to focus on. It is a structural pattern of how the model manages information flow during the encoding process.

## 6 Failure Analysis

Several potential failure modes are present in this implementation:

1. **Overfitting:** With only 500 training samples and near-zero training loss, the model has almost certainly memorised the training set rather than generalised. The consistently elevated test loss in Figure 2 confirms this. Solutions include dropout, weight decay, or training on a larger data split.
2. **Data Hunger:** Transformer-based models generally require substantially more data to learn effective attention patterns. On very small datasets, simpler models may generalize better.